

ACM RECSYS 2026 · RECSYS CHALLENGE WORKSHOP

State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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Minneapolis · 2 October 2026 · Music-CRS track, team npatta01

47,071

tracks in the catalog

11

ways to search

1

learned ranker

A decorative bar chart at the bottom of the slide, consisting of numerous vertical bars of varying heights in a dark gray color, creating a rhythmic pattern across the width of the slide.

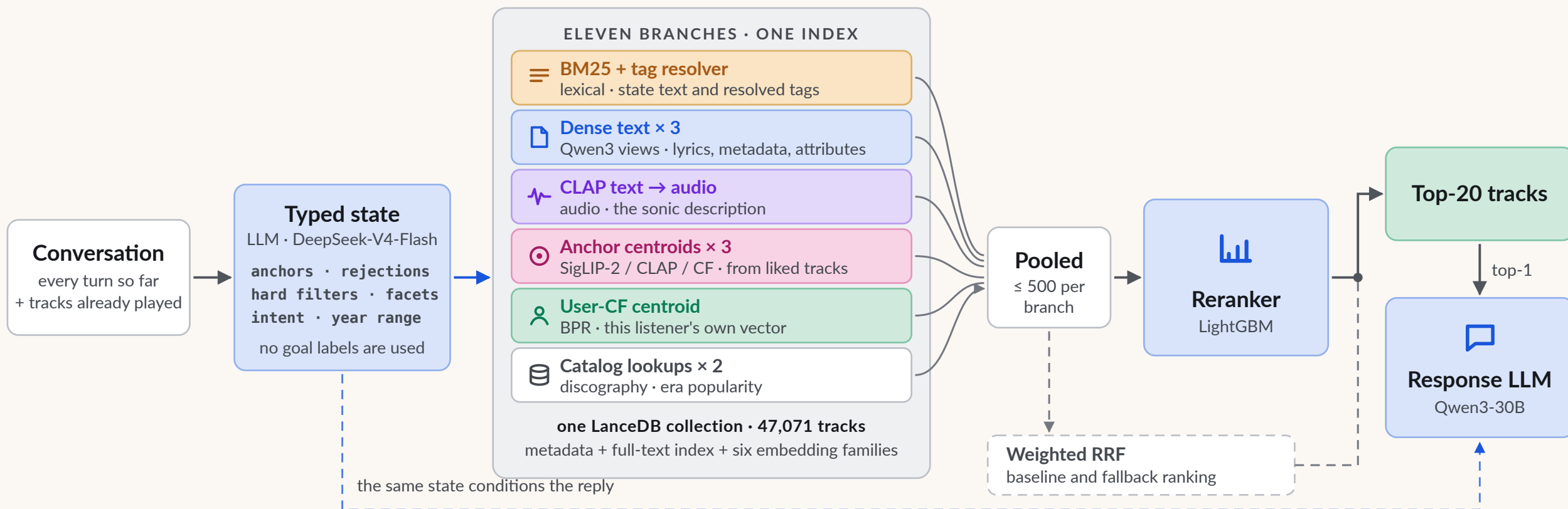


Contents

1	How it works	the state, retrieval, reranking, the reply
2	How it scored	the official results
3	Why we did not score higher	labels, training, ranking, missing data, the reply
4	Conclusion	what we showed, and questions

≡ One state, eleven branches, one learned ranker

APPROACH



Organisers supplied the audio (CLAP), cover-art (SigLIP-2), listening-data (BPR) and small text (Qwen3-0.6B) embeddings; we added Qwen3-8B text embeddings and our own bi-encoder.

Request understanding

APPROACH

WE HAD JUST PLAYED



Ryan Adams & The Cardinals — “Let Us Down Easy”

DeepSeek
V4-Flash



LISTENER

...more tracks with that kind of emotional depth and acoustic focus, maybe branching out to other singer-songwriter or alt-country artists?

CONVERSATION STATE

REQUEST TYPE **new artist**

AVOID **Ryan Adams & The Cardinals**

MOOD **introspective, emotional depth**

SOUND **acoustic-driven**

GENRE **singer-songwriter, alt-country**

DeepSeek-V4-Flash reads **every turn so far** and writes a typed **state**: “other artists” after a Ryan Adams track becomes an artist to avoid, and the mood carries over.



What the state can hold

APPROACH

THE REQUEST

- request type
- intent mode
- routing flags

WHAT THEY LIKE

- anchors and pinned tracks
- per-track feedback
- mentioned entities

WHAT THEY RULED OUT

- rejected tracks
- rejected artists
- hard filters

WHAT IT SOUNDS LIKE

- mood facets
- genre facets
- sonic facets

WHAT IT IS ABOUT

- lyrical theme

BOUNDS

- release-year window
- process constraints



How the state drives retrieval

APPROACH



The request

request type · intent mode · routing flags



Which branches fire, and how much each one weighs in the merge

routing weights

pivot skips anchors

gates discography



What they like

liked tracks · named artists and tags



Search near liked tracks; match named entities

anchor centroids ×3

BM25 fields

artist discography



What it sounds like · is about

mood · genre · sonic · lyrical theme



The text, audio and lyrics queries

dense text ×3

CLAP text→audio

lyrics view



What they ruled out

rejected tracks and artists



Dropped from every branch before the ranker

hard drop

same-artist demote



Bounds

release-year window · date filters



Mask out-of-range tracks; boost the era

candidate mask

era-popularity lookup

Retrieval — eleven branches

APPROACH



×1

BM25 + tag resolver

keywords and resolved tags
from the state



×3

Dense text

Qwen3 embeddings of lyrics,
metadata, attributes



×1

CLAP text → audio

the sonic description, matched
to audio



×3

Anchor centroids

near liked tracks: cover art,
audio, listening



×1

User-CF centroid

this listener's own BPR vector



×2

Catalog lookups

the artist's discography; era
popularity

Each branch returns its **top 500**; the ranker sees their union.



Reranking

APPROACH

Pooled candidates

top 500 per branch



LightGBM ranker

146 features per candidate



Top 20

the scored list

WHAT IT LEANS ON · SHARE OF SPLIT GAIN

Bi-encoder cosine

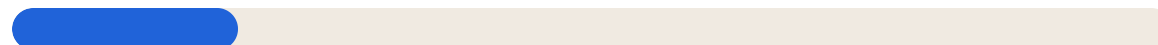
· conversation vs track card



59.0%

Per-branch rank and score

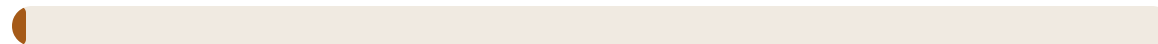
· 66 features



11.5%

Constraints and rejections

· 13 features



0.7%



Response generation

APPROACH

INPUTS

Conversation

"...it feels a bit too serious. I'm looking for something more uplifting and positive..."

Latest state

Mood **uplifting, positive**

Avoid **serious**

Top-ranked track

Track **Röyksopp – "Happy Up Here"**

Tags **cheerful, hopeful, electronic...**



Qwen3-30B

temperature 0 • one
pass

1-2 sentences about
the top track only



OUR REPLY

"Perfect – 'Happy Up Here' by Röyksopp is exactly the kind of uplifting, cheerful electronic track you're looking for. With its bright synths and hopeful groove, it's designed to lift your spirits and make you smile."

✓ How it scored

RESULTS

OFFICIAL CODABENCH SPLIT	NDCG@20	CAT.	LEX.	JUDGE	COMPOSITE
Blind-A · development phase	0.4380	0.0313	0.8028	4.70	0.5799
Blind-B · final	0.2537	0.0315	0.7862	3.30	0.3811

NDCG@20

-42%

0.4380 → 0.2537

JUDGE (1-5)

-1.40

4.70 → 3.30

DIVERSITY TERMS

flat

catalog and reply-wording
diversity barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.



Why we did not score higher

WHAT IS NOT IDEAL

- 1 **Inconsistent labels**
- 2 Why we trained on the dev set
- 3 Extracted, but the ranker did not act on it
- 4 No data from outside the catalog
- 5 Gaps in our response generation

Inconsistent labels — “from a different artist”

WHAT IS NOT IDEAL

RECOMMENDER AGENT PLAYED



Bonobo — “Cirrus”

LISTENER, NEXT TURN

...something with a similar chill, electronic vibe, maybe with a subtle beat,
but **from a different artist?**

ANSWER KEY FOR THAT TURN



Bonobo — “Jets”

MOVES TOWARD GOAL

× Our judges: wrong — same artist

NEXT TURN

“Jets” is cool, but I was actually hoping for something **from a different artist**
this time...

ANSWER KEY



Bonobo — “Pieces”

MOVES TOWARD GOAL

× Our judges: wrong — same artist

17% of training turns keep the artist the listener just asked to leave
Two LLM judges re-scored all 113k labelable turns; Opus broke ties.



Why we trained on the dev set

LESSONS



State extraction: 1 paid LLM call per turn

to read each turn into a state



Too costly to extract state for 121.6k training turns

so no ranker features there



Trained on the 8,000 dev turns

the same turns we checked it on

Dev, turns it trained on



0.384

Blind-B, official



0.254

nDCG@20. The dev row is an earlier model version, shown because it is the number we selected on. Source: paper §4.3, §5.3.



Extracted, but the ranker did not act on it

WENT WRONG

"I gotta break away from Deltron and Del completely"	✓ STATE EXTRACTED avoid Deltron 3030, Del	✗ OUR TOP 20 10 of 20 by them — from rank 6
"branching out to other singer-songwriter artists"	✓ STATE EXTRACTED avoid Ryan Adams	✗ OUR TOP 20 10 of 20 by Ryan Adams — rank 1
"still focusing on Juanes... haven't found the depth I'm after"	✓ STATE EXTRACTED rejected Juanes	✗ OUR TOP 20 7 of 20 by Juanes — rank 1



No data from outside the catalog

WHAT IS NOT IDEAL

A tempo and a key

“exactly 126.70 bpm, in G minor”



the catalog has no tempo or key field
no branch can match on either

A soundtrack

*“a specific 80s synth-pop song from the **Breaking Bad** soundtrack”*



we served Soft Cell — “Tainted Love”
no film or TV links in the catalog

The state was right both times. The gap is **knowledge, not understanding**.



Gaps in our response generation

WHAT IS NOT IDEAL

The track is not in the catalog, so retrieval could not return it — the reply presents our substitute as if it were the request

1

"I wanted 'Watercolors' by Pat Metheny specifically."

"Got it — here's 'Alfie' by Pat Metheny..."

NOT FLAGGED

Our pick broke the listener's constraint — the reply recommends it anyway

2

*"...something from a **different artist** this time"*

"...while this track is by him, it's from a different era and band..." — Ryan Adams again

NOT FLAGGED

One pass, no check: the reply could flag **neither unavailable tracks nor unmet constraints**. The prompt was chosen on Blind-A (judge 4.70 there, 3.30 on Blind-B).



Conclusion

WHAT WE SHOWED



A typed conversation state

- one LLM call per turn
- request, likes, rejections, sound, bounds



Retrieval → ranking → reply

- eleven retrieval branches
- a learned ranker
- a reply about the top track



Gaps in our system

- inconsistent labels — 17% keep a rejected artist
- trained and checked on the same turns
- the ranker did not act on the state
- no knowledge outside the catalog
- the reply was never checked

Thank you – questions?



Paper, poster, slides, code, models, data

npatta01.github.io/music-crs-2026

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→ Appendix

TYPE A NUMBER + ENTER · OR G TO SEARCH

- | | | | | | |
|--------------------|----------------------------|--------------------|--|--------------------|--|
| 1 | Where our points came from | 2 | The task | 3 | Glossary |
| 4 | State extraction | 5 | Six request types, six shapes of state | 6 | Routing flags decide which branches fire |
| 7 | Entity resolution | 8 | The eleven branches | 9 | Fusion and fallback |
| 10 | b1 training | 11 | Development-split results | 12 | Failure 3 — the better track was already in the pool |
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| 16 | Related work | 17 | Reproducing the run | 18 | The target often ignores the request |
| 19 | The reranker | 20 | Down-weighting contradicted turns | 21 | The b1 bi-encoder |
| 22 | How the relabelling worked | 23 | What the relabelling found | | |

Where our points came from

RESULTS

$$0.50 \cdot \text{nDCG@20} + 0.10 \cdot \text{cat.div} + 0.10 \cdot \text{lex.div} + 0.30 \cdot (\text{judge} - 1)/4$$

OUR BLIND-B DECOMPOSITION

0.50 × nDCG@20	0.2537	0.1269
0.10 × cat. div.	0.0315	0.0032
0.10 × lex. div.	0.7862	0.0786
0.30 × (3.30-1)/4		0.1725
Composite		0.3811

WHERE THE SCORE CAME FROM

The **judge term** out-earned **nDCG**, and catalog diversity returned **0.003** of the 0.10 on offer. We optimised almost entirely for relevance and left the diversity terms alone.

EVERY ASSISTANT TURN

- Retrieve **20 tracks** from a shared **47,071-track** catalog
- Write a short reply presenting the top recommendation

PER-TRACK METADATA

Title, artist, album, a median of **17** genre/mood tags, popularity, release date, plus precomputed text, audio, cover-art and collaborative-filtering embeddings.

SPLITS

Train	15,199 sessions (~121.6k turns)
Development	1,000 sessions × 8 turns = 8,000
Blind-A / Blind-B	80 sessions each, final turns

Blind-B additionally withholds the conversation-goal annotations present in training data.

BUILT ON

TalkPlayData 2. The conversations are **LLM-generated**, not observed listening sessions — which matters later.



Glossary

BACKGROUND

Typed state

Structured fields an LLM reads off the conversation each turn: artists to anchor on, rejected tracks, filters, mood.

Dense embedding

A vector from a neural model; texts with similar meaning land close together.

SigLIP-2

The same idea for text and images — here, album covers.

Centroid search

Average the vectors of tracks the listener liked, then search near that average.

LambdaMART · LightGBM

Gradient-boosted decision trees trained to order a list; LightGBM is the library.

LLM judge

An LLM that rates the reply 1–5; the organisers run it.

BM25

A classic keyword-search score over text fields.

CLAP

A model that puts text and audio in one space, so a description can find a sound.

Collaborative filtering (BPR)

Vectors learned from who listened to what; BPR is the training method.

RRF

Reciprocal-rank fusion: merge ranked lists by summing $1 / (k + \text{rank})$ per track.

Bi-encoder

One model encodes the conversation and the track separately; relevance is the cosine of the two vectors.

In-sample vs out-of-fold

Scoring on turns the model trained on, versus on held-out folds. In-sample overstates.



State extraction

APPROACH

THE EXTRACTOR

DeepSeek-V4-Flash (build 0423), schema-constrained, given the dialog so far and the IDs of tracks already played.

DELIBERATELY BLIND

No conversation goals, goal progress, or user profiles reach the extractor. The state is whatever an LLM can read off the conversation.

WHY THIS MODEL

Cheaper extractors under-filled the optional fields — rejections and facets especially — so every final run uses this one. Per-turn extraction is a paid API call, which is what kept us off the training split.

PROMPT VERSION

ConversationStateV1, projected to V0Plus for the downstream contract.



Six request types, six shapes of state

APPROACH

REQUEST TYPE	A REAL TURN	WHAT THE EXTRACTOR PULLED OUT
<code>exact_track</code>	"I wanted <i>Watercolors</i> by Pat Metheny specifically."	track = <i>Watercolors</i> must_use · artist = Pat Metheny must_use
<code>exact_album</code>	"Which song from Peter Doherty's <i>Grace/Wastelands...</i> "	artist + album must_use · facet "best tracks of 2009"
<code>same_artist</code>	"Got another track known for his guitar work?"	artist must_use · prior track do_not_use · facets: guitar work, melodic complexity
<code>new_artist</code>	"...branching out to other singer-songwriter artists."	artist = Ryan Adams & The Cardinals do_not_use · facets: acoustic, introspective
<code>attribute_search</code>	"...a really striking, abstract cover art, bold colors and shapes."	three visual facets · routing flag image_or_visual_search
<code>hidden_target</code>	"A Lil Wayne song from an experimental phase."	artist must_use · facet "experimental" · flag hidden_target_search

The extractor read all six of these correctly. Across the whole Blind-B run only **3 of 80 turns** were diagnosed as a state-extraction miss — reading the conversation was almost never the problem.



Routing flags decide which branches fire

APPROACH

FLAG	OPENS
exact_entity_probe	catalog lookups — discography, album
lyric_search	the Qwen3 lyrics view
feature_articulation	CLAP text→audio on the sonic description
image_or_visual_search	the gated SigLIP-2 cover-art branch
hidden_target_search	widens the pool; suppresses exact pins

A REAL VISUAL REQUEST

“I'm looking for an album with a really striking, abstract cover art... bold colors and shapes, almost like a modern painting.”

→ three visual facets, `image_or_visual_search = true`

AND THE HONEST PART

The reranker carries **no features** for that branch. Even when the flag opens it, the ranker cannot weigh what it returns.



Grounding names to the catalog

APPROACH

NAMES → IDS

Fuzzy string matching of artist and track names against the catalog name lists, keeping matches above **80/100**. Downstream stages only ever see identifiers.

FREE TEXT → TAGS

- 1 Exact match against the tag vocabulary
- 2 Alias match
- 3 Substring match
- 4 Embedding nearest neighbour, cosine ≥ 0.6 , top 3

Used by the BM25 branch. Outside it, tag matching falls back to exact strings — part of why so little of the state became executable.

The eleven retrieval branches

APPROACH

BRANCH	SPACE	QUERY
BM25 (fielded)	lexical	state text + resolved tags
Dense text × 3	Qwen3 views	state-built strings
CLAP text→audio	audio	sonic description
Anchor centroids × 3	SigLIP-2/CLAP/CF	liked-track centroids †
User-CF centroid	CF-BPR	the listener's own vector
Lookups × 2	catalog	discography; era popularity

GATED TWELFTH

SigLIP-2 text→cover-art fires only on a visual request. The reranker carries no features for it.

POOL DEPTH

Each branch truncated to its top 500 before the union reaches the ranker.

† Gated by intent mode: a pivot suppresses anchor centroids.

Appendix

B — back to the talk · A — index · number + Enter — jump



Weighted RRF — baseline and fallback

APPROACH

FUSION

Weighted reciprocal-rank fusion, $k = 60$, top-1000 across the branch pools.

POST-FUSION ADJUSTMENTS

- Drop rejected artists and tracks
- Drop tracks already played
- Promote / demote on exact tag membership
- Year-range decay

AS A BASELINE

0.1492

devset nDCG@20, no training

STILL THE FALLBACK

If the learned ranker is unavailable, this order ships. It is the only part of the ranking stack with no trained component.



How b1 was trained

APPROACH

$$L = -(1/B) \sum_i \log \left[\frac{\exp(v_{q_i}^T v_{d_{i+}} / \tau)}{\sum_{d \text{ in } D_i} \exp(v_{q_i}^T v_d / \tau)} \right] \quad \tau = 1/20$$

POSITIVES

53,885

(turn, track) pairs whose *next* turn's goal-progress says the recommendation moved the session forward

NEGATIVES

4 + batch

four mined hard negatives per positive, plus every in-batch document

RETRIEVAL VS RANKING

As a twelfth *retrieval* branch it added little to the union's coverage and was dropped. As a *ranking* feature it dominates the fit — out-of-fold nDCG@20 0.1970 → 0.2032.

Not isolated: which parts of the rendering matter, and whether the effect carries to the blind splits.



Development split — what we steered on

RESULTS

SYSTEM (HOW MEASURED)	NDCG@20
Weighted RRF fusion (no training)	0.1492
LambdaMART, out-of-fold CV	0.1970
+ b1_cos (deployed feature set)	0.2032
Historical v10 LambdaMART, in-sample	0.3844
Official Blind-B	0.2537

DO NOT OVER-READ THE FIRST GAP

The fusion and learned-ranker configurations used **different candidate-pool depths**, so 0.149 vs 0.197–0.203 does not isolate the effect of learned re-ranking.

THE IN-SAMPLE ROW

A historical goal-aware model without b1_cos.
Not directly comparable — shown because it is the number we were selecting on.



Failure 3 — the better track was already in our pool

ANALYSIS

WHAT THEY ASKED FOR	WE SERVED AT RANK 1	ALREADY IN OUR POOL	POOL RANK
nu-jazz, downtempo, female vocals — <i>not</i> Parov Stelar	Parov Stelar — Booty Swing	St Germain — What You Think About...	34
new underground hip-hop, not Deltron	Dead Prez — Hip Hop	Souls Of Mischief — Make Your Mind Up	31
hardcore, not classic hip-hop	Dead Prez — Hip Hop	Papa Roach — Getting Away With Murder	49
that carol, but not Beegie Adair again	Beegie Adair — The Christmas Song	Frank Sinatra — Have Yourself A Merry Little Christmas	63

On **31 of 80 Blind-B turns** a non-violating, better-fitting candidate was sitting in our own pool below the top-20 we served. Retrieval found it; ranking buried it.

Auditing the submitted run

ANALYSIS

SINGLE-JUDGE QUALITY RATING

78%

of Blind-B turns rated weak-or-bad by one DeepSeek-V4-Flash judge

READ AS

Diagnostic, not ground truth. Blind-B exposes no relevance labels, so only the validation failure is actually *measured*.

DETERMINISTIC TRACE AUDIT — 43 DIAGNOSES OVER 32 TURNS

Heuristic alternative below top-20	31
Under-applied constraint	8
State-extraction miss	3
Entity-resolution miss	1

Categories overlap. The alternative was chosen by positive-term metadata matches plus popularity — an audit heuristic, not a relevance judgment.



Limits of our own analysis

ANALYSIS

- The relabeling was **never checked against human judgments**.
- The Blind-B quality audit used a **single** LLM judge.
- The trace audit was **heuristic**, not a relevance judgment.
- The conversations are **LLM-generated**, not observed listening sessions.
- Neither blind split exposes relevance labels — blind-set failure rates and relabeling rates are **diagnostic, not ground truth**.
- Development-split and leaderboard numbers are **separate measurements**.



What we would do differently

CONCLUSION

- 1 **Hold out a clean split before touching anything.** Free, and it would have changed every selection decision we made.
- 2 **Make the state executable.** Rejections and constraints carried 0.7% of model gain — most of what the extractor understood never became a filter.
- 3 **Import knowledge the catalog lacks** — tempo, key, soundtrack and film/TV links — so requests naming them have something to match.
- 4 **Check the response against the state** before emitting it: flag unavailable tracks and unmet constraints instead of overselling a substitute.
- 5 **Decide what to do when the benchmark and your judgment disagree.** We still do not have a principled answer.

→ Related work — four established lines

A1

CONVERSATIONAL STATE TRACKING

CRS systems commonly track dialog state explicitly and condition retrieval on it. We follow that, but compile the state with a **prompted LLM** instead of a trained tracker.

MODALITY-SPECIFIC MUSIC ENCODERS

Contrastive language-audio pretraining (CLAP), vision-language over cover art (SigLIP-2), collaborative filtering (BPR), general text embeddings (Qwen3).

RETRIEVE-THEN-RERANK AT CATALOG SCALE

Pool multiple sources with reciprocal-rank fusion; rank with LambdaMART.

OUR CLAIM

What we report is **how that combination behaved** — not a new technique.



Reproducing the run

A2

FROZEN REPLAY – NO CREDENTIALS

Source the offline bundle's env and every LLM call is served from a curated cache. A miss hard-fails rather than silently spending money.

```
source .repro/scripts/activate_repro_env.sh
```

LIVE RE-RUN

```
python run_experiment.py --backend modal --tid state_ranker_v10_lgbm_blindset_B --eval_dataset blindset_B
```

ARTIFACTS

Models and data: [hf.co/datasets/Npatta01/music-crs-repro-2026](https://huggingface.co/datasets/Npatta01/music-crs-repro-2026)



The target often ignores the request

ANALYSIS

LISTENER · JUST PLAYED BONOBO

...something with a similar chill, electronic vibe, but **from a different artist?**

ANSWER KEY



Bonobo — “Jets”

MOVES TOWARD GOAL

NEXT TURN

“Jets” is cool, but I was hoping for something **from a different artist this time...**

ANSWER KEY



Bonobo — “Pieces”

MOVES TOWARD GOAL

The answer key repeats the artist the listener just asked to leave. Our ranker **anchors** the same way — we think it learns it here.



What the ranker leans on

APPROACH



Share of the ranker's split gain, by feature family. One signal — the bi-encoder cosine — carries **59%**; constraints and rejections carry **0.7%**.



Down-weighting contradicted turns

APPROACH

Every turn starts at		1.0
Next turn rejects the artist		× 0.6
Next turn rejects the track		× 0.3
Goal note: did not help		× 0.3
Never below		0.2

WHY

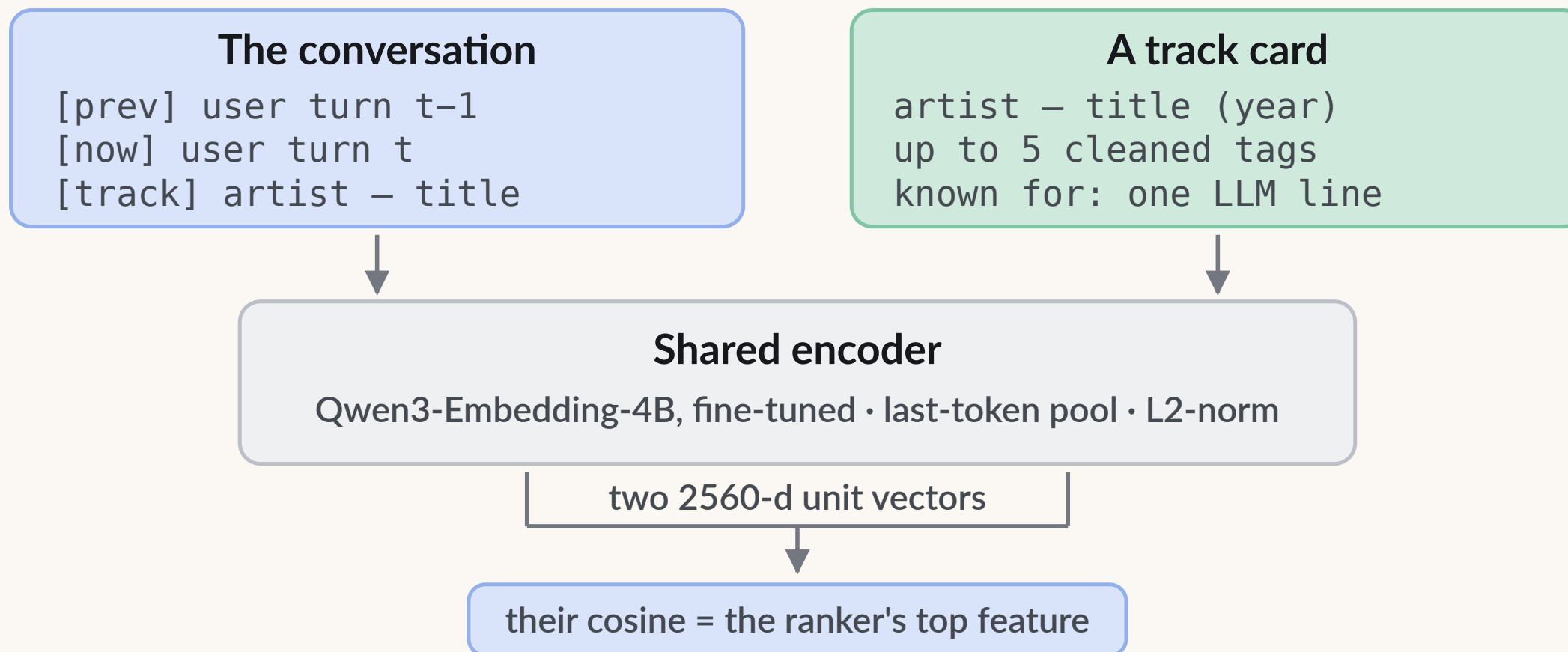
The label is whatever played next. When the **next** turn says it was wrong, that label counts for less in training.

Chosen by hand, never tuned. Goal notes shape these weights but are never a ranker input.



The bi-encoder behind that top feature

APPROACH



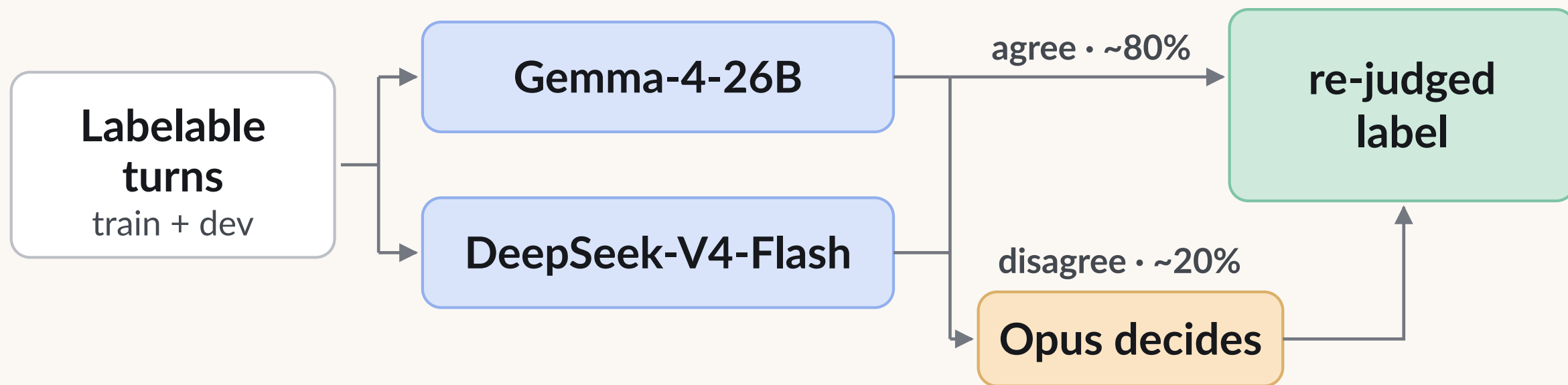


How the LLM relabelling worked

ANALYSIS

The answer keys come from the simulated recommender's picks. To see how often they actually fit the request, we had **two LLM judges re-score every labelable turn**.

Both judges score the same two axes: was a different artist asked for, and does the track fit the request?



Agreement split measured on the training turns.



What the relabelling found

ANALYSIS

17%

of training turns keep the artist the listener had just asked to leave

18,222 turns — both judges or the arbiter marked the label as artist anchoring

113,393

turns re-judged

every labelable train and dev turn

Retrained on these labels, the ranker scored lower **against the original labels** — a test that could never favour the fix — so we kept the originals.